

Trust-Based Task Offloading Framework for Edge-Based IoT Systems

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Abstract—The Internet of Things (IoT) has led to a shift of localized sensing and actuation to the network fringe. Localized sensing and local actuation is an emerging phenomenon at the network fringe, thanks to the Internet of Things (IoT). However, the limitations in terms of size and resources in IoT devices necessitate the transfer of complex tasks to intermediaries edge servers. While there are substantial advantages for scalability in regard to latency by moving the application closer to where data is collected—from cloud to edge—the sheer number of edge nodes offers significant challenges: Edge nodes are very diverse, and hardware reliability, network topology and level of security vary significantly. The traditional load-balancing schedulers distribute the tasks to the nodes based on proximity or the load on the CPU, without taking any account of what the data indicates, assuming that the nodes are honest. Appropriately, we suggest a complete Trust Based Task Offloading Framework (TBTOF) to calculate dynamically the trustworthiness of the edge nodes. We're employing a multi dimensional weighted formula that factors in task success percentage, CPU availability, response time, and historical uptime to generate a current trust score. The framework treats problematic nodes (blacklisted, mechanically failed) as if they are malicious and actively quarantines them, with a very strict trust threshold. We mathematically show the stability and whiteness of the algorithm using the approach of a discrete time markov chain. The results from a system evaluation using a Python-based simulation environment with a naive randomized baseline show a 22% increase in the overall task success rate and an 28.5ms decrease in average processing delay. Furthermore, the energy savings for the networks as indicated by EDP and the CDF results, further support the fact that this framework has a great impact on energy savings. The architecture provides Rising to the challenges of future scenarios such as Vehicular Edge Computing and the Internet of Medical Things, it is extremely transparent, auditable and extensible.

Index Terms—Edge Computing, Internet of Things, Trust Management, Task Offloading, Security, Resource Allocation, Markov Chains

I. INTRODUCTION

The move to the "edge" from centrally located cloud-based computing is a change in today's Internet architecture. New opportunities in smart city infrastructures, the implementation of the network of the Industrial Internet of Things (IIoT) and the emergence of autonomous vehicular networks have generated an enormous amount of real-time data that must be processed and also localized in real time [1]. However on the other side of the coin, the IoT devices have very restricted processing capacities, memory and battery life. These problems are solved by having edge servers as intermediaries between the data source and the cloud, where an edge server

with heavy workloads of computations is used. The solutions to these problems lie in the concept of edge computation, in which the edge servers are essentially used as intermediate servers or nodes, and are mainly located near the data sources geographically, to answer heavy workloads of computation.

Edge computing paradigm's ROI in terms of reduced latency and lower core network bandwidth usage comes with some security and reliability issues. Edge networks span a highly varied hardware stack of from fully-resourced telecom micro-data centers to very resource-constrained, semi-public edge devices – versus the levels of control and monitoring found in centralized cloud data centers. The high degree of decentralization greatly increases the threat surface, opening the door to physical tampering, hardware degradation, and the meaningful packet dropping, task hoarding, as well as malicious behaviors in the network [3].

Contemporary orchestration and routing paradigm considers the edge environment as a single system. The current schedulers are mostly based on local measures like the proximity of the destination node or immediate CPUs or load balancing, and they do not consider the previous reliability or behavioral integrity of the destination node. This means that, with blind task allocation, in case an edge node is breached via attacks like Sybil for instance, or if hardware in such node fails silently, loss of packets, privacy violation, and significant deadline failures are faced [4].

To tackle these vulnerabilities, a paradigm shift in edge orchestration systems is needed, in which 'trust' becomes the first-class and mathematically rigorous routing factor. The main goal of this work is to propose a multi-dimensional Trust-based Task Offloading Framework (TBTOF) which continuously assesses trustworthiness of edge nodes before offloading computational tasks. The proposed framework combines task success rate, computational capacity, latency and historic availability into an integrated trust metric to improve the reliability, efficiency and security in the delegation of tasks among IoT devices configured in a heterogeneous edge computing environment.

A. Problem Statement

Task offloading in DECNs is complex in both aspects. Their operational reliability is significantly different from each other because they are equipped with different types of hardware, encountering unpredictable resource contention and varying

workloads, as well as dynamic network conditions [5]. The edge environments generally have different devices that vary across their computation power, availability, stability, etc., which leaves the reliability of executing tasks challenging.

Second, most existing orchestration and scheduling systems lack integrated trust evaluation mechanisms. Current task allocation strategies primarily focus on metrics such as proximity, latency, or instantaneous CPU availability, while ignoring the historical behavior and integrity of participating nodes. In the absence of continuous trust assessment, IoT applications are forced to rely on potentially unreliable or compromised edge nodes for computation.

This operational blind spot enables malicious, malfunctioning, or poorly performing nodes to attract and retain computational tasks despite exhibiting unreliable behavior. Consequently, blind task offloading can lead to packet loss, increased processing delays, degraded Quality of Service (QoS), compromised data privacy, and reduced overall network integrity. Therefore, there is a critical need for a lightweight and mathematically grounded trust-aware task offloading framework capable of dynamically identifying and isolating untrustworthy edge nodes while ensuring reliable and efficient computation in heterogeneous IoT-edge environments.

B. Contributions

To address the risks associated with blind task offloading in heterogeneous edge environments, this paper proposes a comprehensive Trust-Based Task Offloading Framework (TBTOF) with the following major contributions:

- We design a dynamic multi-dimensional trust model (τ) that evaluates edge nodes across four orthogonal parameters: task success rate, CPU availability, response latency, and historical uptime stability.
- We formalize a detailed threat model covering malicious and unreliable node behaviors, and employ a discrete-time Markov chain formulation to mathematically analyze the stability of the trust adaptation process and its resistance to whitewashing attacks.
- We develop an interpretable and auditable decision-making engine that enforces a strict trust threshold ($\tau_{\text{threshold}} = 0.55$) to isolate malicious or unreliable nodes while ranking trusted nodes using a hybrid resource-trust utility function.
- We validate the proposed framework by extensive simulations for various network conditions and adversarial scenarios. Results from experimental tests show great improvement in system reliability and efficiency both in terms of Energy-Delay Product (EDP) analysis and in terms of latency Cumulative Distribution Function (CDF).
- Simulation results show that the proposed framework achieves approximately a 22% improvement in overall task success rate and substantially reduces processing latency compared to conventional randomized baseline schedulers.

II. RELATED WORK

The problem of secure and efficient task offloading has driven significant advancements in edge computing research, shifting the focus from conventional mathematical optimization techniques toward security-first and trust-aware paradigms.

A. Trust-Aware and Multi-Feedback Architectures

Initial solutions to network edge management issues were seen as balancing security and latency. Our framework is inspired by the early multi-feedback trust model by Kong et al. [6] that integrated the task success rate and latency, forming the basic framework on which all existing models were built. Recently, Chen et al. [7] showed that a goal-driven collaborator selection mechanism can adapt dynamically to the changes of network status by using collaborator trust weight which primarily depends on application and payload needs.

Moreover, Nitti et al. in [8] proposed formal behavioral trust vectors to identify unreliable resource providers that possess reliable word-of-mouth high computational capability but are unable to provide an acceptable computational capability. All of these studies emphasize the increasing relevance of including a behavioral trust measure in edge orchestration system.

B. Reinforcement Learning and Intelligent Scheduling

Deep Reinforcement Learning (DRL) has emerged as a dominant approach for dynamic task scheduling and resource allocation in edge environments. Li et al. [9] developed a reliability-aware scheduler for cloud-fog computing systems using an Advantage Actor-Critic (A2C) algorithm. Similarly, the QLB-IoT framework [3] integrated Q-learning with blockchain technology to optimize energy consumption while enabling adaptive traffic routing in IoT-edge networks.

In Vehicular Edge Computing (VEC), Deep Deterministic Policy Gradient (DDPG) algorithms have been employed to address sparse roadside infrastructure coverage and highly dynamic vehicular mobility patterns [10]. Although these learning-based approaches provide strong theoretical optimization capabilities, they often suffer from limited interpretability, high computational overhead, and the need for extensive training epochs.

In contrast, our framework intentionally avoids black-box machine learning models in favor of a lightweight and deterministic trust computation mechanism that is fully interpretable and suitable for resource-constrained IoT environments.

C. Blockchain and Cryptographic Integration

For enterprise-scale and highly secure deployments, researchers have increasingly adopted blockchain and distributed ledger technologies for trust management. Zuo et al. [11] demonstrated how decentralized ledgers can be used to permanently record node reputations in mobile crowdsensing systems. Building upon this concept, the PAIS 2024 framework

[12] integrated blockchain with Digital Twin (DT) architectures to support secure vehicular edge computing environments.

At the extreme edge, the TUB-IoT architecture [13] introduced quantum-resilient trust mechanisms for security-centric IoT deployments. While these blockchain-based approaches provide strong security guarantees and immutability, they often incur significant consensus latency and computational overhead. Consequently, lightweight threshold-based trust models such as the one proposed in this work remain highly competitive for resource-constrained IoT-edge environments where low latency and energy efficiency are critical requirements.

III. SYSTEM ARCHITECTURE AND DESIGN

The proposed framework adopts a three-tier edge computing architecture consisting of highly resource-constrained IoT devices at the network edge, an intermediary layer of edge servers responsible for primary computation, and a distant cloud layer that functions as a fail-safe storage and backup repository. This hierarchical architecture enables low-latency processing while preserving scalability and reliability in heterogeneous IoT environments.

A. System Model

Let the edge network be formally represented as a set of active edge nodes:

$$\mathcal{N} = \{EN_1, EN_2, \dots, EN_n\} \quad (1)$$

At any discrete time instant t , each edge node is characterized by a state vector representing its operational health and computational reliability:

$$S_i(t) = \langle C_i, L_i, SR_i, A_i \rangle \quad (2)$$

where:

- C_i denotes the available CPU capacity percentage of node EN_i ,
- L_i represents the average task processing latency in milliseconds,
- SR_i indicates the historical task success rate, and
- A_i represents the uptime/availability of the infrastructure of the node.

At the same time, the network is fed with a stream of computations:

$$\mathcal{T} = \{T_1, T_2, \dots, T_m\} \quad (3)$$

The Trust Orchestrator performs a four-stage decision pipeline for each task that arrives [15]:

- 1) **Real-Time Data Collection:** The orchestrator continuously polls edge nodes to gather current hardware and historical performance metrics.
- 2) **Trust Computation:** The collected state vector $S_i(t)$ is processed using the proposed multi-dimensional trust evaluation formula to compute the trust score τ_i for each node.

- 3) **Decision Engine:** Nodes with trust scores below a predefined threshold are filtered out, while the remaining candidate nodes are ranked according to a hybrid utility function.
- 4) **Dynamic Trust Update:** After task execution, the trust score of the selected node is updated dynamically based on the observed execution outcome and post-execution system state.

IV. FORMAL THREAT MODEL

To rigorously evaluate the security posture of the framework, we define a formal threat model encompassing the behavioral anomalies specifically mitigated by our architecture.

A. On-Off Attacks (Whitewashing)

In whitewashing attack, a compromised node pretends to be trustworthy for a long time to boost its trust value τ_i to high levels, and then it tricks the processing core into dropping critical payloads when a high value task comes from it. This is a problem that traditional averaging algorithms are extremely vulnerable to. The asymmetrical update protocol (Section VII.D). If an attacker fails, they lose 0.045 but on a successful roll they only add 0.025. If the node is whitewashing, it will shortly fall below the strict 0.55 limit and will be permanently banned from going through a damaging cycle.

B. Bad-Mouthing and Ballot Stuffing

Weak nodes could be malicious and send false false report for honest nodes (Bad-Mouthing) or false success reports for the compromised nodes (Ballot Stuffing). We have integrated into the model the direct measurement of latency L_i and the computation execution, which is done through the central Trust Orchestrator and hence avoids the local collusion problem that occurs in the peer-to-peer approach.

C. Denial of Service via Resource Exhaustion

A node may attempt to post an artificially high CPU set available ($C_i = 100\%$) so that other nodes will not take up any jobs which will lead to bottlenecks in the local node. The CPU weight is limited to $\beta = 0.25$ and the historical throughput is assigned a higher weight ($\alpha = 0.35$), which means that nodes which are not able to satisfy past requirements will be assigned a weight of zero.

V. MATHEMATICAL MODELING OF TRUST STATE TRANSITIONS

In order to formally validate the stability of the proposed trust adaptation mechanism, we model the trust evaluation process as a discrete-time Markov chain. Each edge node is associated with a trust level that can be classified in one of three macroscopic states:

- **Trusted (S_T):** $\tau \geq 0.75$
- **Suspicious (S_S):** $0.55 \leq \tau < 0.75$
- **Quarantined (S_Q):** $\tau < 0.55$ (Absorbing State)

Suppose the transition probability matrix for a node trying to manipulate the system with occasional failure of tasks is P . Following the asymmetric rules for updating trust:

$$\Delta_{\text{success}} = +0.025 \quad (4)$$

$$\Delta_{\text{fail}} = -0.045 \quad (5)$$

the transition matrix is defined as:

$$P = \begin{bmatrix} p_{T \rightarrow T} & p_{T \rightarrow S} & p_{T \rightarrow Q} \\ p_{S \rightarrow T} & p_{S \rightarrow S} & p_{S \rightarrow Q} \\ 0 & 0 & 1 \end{bmatrix} \quad (6)$$

The state S_Q is an absorbing state because once the trust score of a node falls below the threshold value of 0.55, the node receives no additional tasks. Consequently, it cannot generate successful execution events ($E_{\text{result}} = 1$) to increase its trust score. Therefore:

$$p_{Q \rightarrow Q} = 1 \quad (7)$$

which permanently isolates malicious or unreliable nodes from the network.

Furthermore, the expected number of contiguous malicious actions required to permanently quarantine a fully trusted node ($\tau = 1.0$) can be calculated as:

$$k = \left\lceil \frac{1.0 - 0.55}{0.045} \right\rceil = \lceil 10.0 \rceil = 10 \quad (8)$$

This mathematically demonstrates that a whitewashing node can drop at most 10 consecutive tasks before being permanently removed from the edge topology, thereby significantly limiting the effective blast radius of the attack.

VI. COMPUTATIONAL AND OVERHEAD ANALYSIS

In practical edge environments, battery limitations and thermal constraints require the orchestrator's decision-making mechanism to maintain extremely low computational and communication overhead.

A. Time and Space Complexity

During Stage 1 (Real-Time Collection) and Stage 2 (Trust Computation), the orchestrator processes the state vector:

$$S_i(t) = \langle C_i, L_i, SR_i, A_i \rangle \quad (9)$$

for each of the N edge nodes.

Each node requires only four floating-point operations, resulting in a computational complexity of:

$$O(N) \quad (10)$$

In Stage 3 (Threshold Filtering and Utility Computation), the orchestrator evaluates:

$$\tau_i \geq \tau_{\text{threshold}} \quad (11)$$

and computes the utility function:

$$U_i \quad (12)$$

for all valid nodes. Determining the node with maximum utility through linear search also requires:

$$O(N) \quad (13)$$

time complexity.

Therefore, the overall computational complexity of the task offloading decision process remains strictly linear:

$$T(n) = O(N) \quad (14)$$

The memory requirement per node consists of four scalar variables, producing a total space complexity of:

$$O(N) \quad (15)$$

This lightweight complexity enables deployment on resource-constrained gateway hardware such as Raspberry Pi edge controllers.

B. Network Communication Overhead

One of the major concerns in dynamic trust-based systems is the communication overhead introduced by continuous metric polling.

In the proposed framework, each edge node periodically transmits the state vector:

$$S_i(t) = \langle C_i, L_i, SR_i, A_i \rangle \quad (16)$$

Assuming standard 32-bit IEEE 754 floating-point representation, the payload size equals:

$$4 \times 4 = 16 \text{ bytes} \quad (17)$$

Including standard UDP/IPv4 headers (28 bytes), the total packet size becomes:

$$44 \text{ bytes/node/cycle} \quad (18)$$

For a dense urban deployment consisting of 500 edge nodes polling at a frequency of 1 Hz, the total network overhead is:

$$O_{\text{net}} = 500 \times 44 = 22000 \text{ bytes/sec} \quad (19)$$

$$O_{\text{net}} \approx 22 \text{ KB/s} \quad (20)$$

This communication overhead is negligible for modern IEEE 802.11ac and 5G edge infrastructures, confirming that continuous trust evaluation does not significantly burden the network it aims to optimize.

VII. REAL-WORLD CASE STUDIES

A. Vehicular Edge Computing (VEC)

An application of Vehicular Edge Computing (VEC) scenario in a dense urban intersection [18] is considered, for instance, to show the potential of the proposed mathematical model in practical applications. Then, the autonomous vehicle finally generates 4K video streams that are streamed in real-time to access its surroundings—including obstacles—using the LiDAR sensors. To reduce the computational load on the onboard devices and practices load of nearby Roadside Units (RSUs) – usually the edge servers – to perform alike detection.

The AV will be able to offload a detection frame merely due to CPU capacity advertising or resource usage of an RSU in a conventional load balancing approach. If the selected RSU is not reliable, structurally damaged or malicious, they cannot receive or send the detection frame in a given time period. In these situations, these delays could lead to the potential for disastrous on-road accidents.

The proposed Trust Based Task Offloading Framework involves the orchestrator checking the trustworthiness of the nearby RSUs before offloading tasks and keeping checking the trustworthiness of RSU whether or not it influences in task offloading. Before the load is allocated, it dynamically analyzes the availability, reliability, computational stability and latency of each node. In the case of an RSU which is considered mathematically unstable and/or untrustworthy, the orchestrator gets rid of it from the workload candidate list as soon as possible and moves it to the mathematically trusted RSU. It can provide operational safety, reliability, and real-time response in an autonomous driving system in a proactive manner using trust knowledge.

B. Internet of Medical Things (IoMT)

The proposed framework goes beyond latency optimization to tackle significant security and privacy issues in healthcare settings. Wearable pacemakers, continuous glucose monitors, and other medical devices that sense data generate highly sensitive telemetry data in an IoMT deployment that needs to be processed locally, preferably in real time, to maintain patient privacy and continue the clinical monitoring process [14].

When an edge server on a hospital network is compromised by malware or is beginning to act in an unstable way, it might appear to be less available, lose more packets, or fail to perform tasks as expected. In the proposed model, this behavioural deterioration is directly manifested by the asymmetric adaptation mechanism of trust, during which the punishment is applied in case of a failed execution event:

$$\Delta_{\text{fail}} = -0.045 \quad (21)$$

The patient data stream is automatically quarantined if the node trust score falls below the specified threshold and the Trust Orchestrator quarantines the server that has been compromised. This guarantees seamless monitoring, and complies with stringent healthcare regulatory standards like the Health

Insurance Portability and Accountability Act (HIPAA). As a result, the framework has reduced the chances of life-threatening service disruptions and will removes service-disrupting broken infrastructure from critical medical data processing workflows.

VIII. IMPLEMENTATION AND EXPERIMENTAL METHODOLOGY

We built up the structure in two complementary modes. The back-end is a pure-functional Python simulator based on explicit state management to simulate multi-node topologies. On the frontend you will find live line charts showing node degradation at 1Hz, as well as an interactive dashboard built with java script.

TABLE I
SIMULATION ENVIRONMENT PARAMETERS

Parameter	Value / Distribution
Total Tasks ($\mathcal{T}$)	10,000 discrete payload requests
Number of Edge Nodes (N)	Variable (6 to 500 nodes)
Payload Size	Uniformly distributed [1MB - 15MB]
Required CPU Cycles	Normal distribution $\mathcal{N}(500, 50)$ Megacycles
Baseline Network Latency	5ms to 12ms (Wireless Channel)
Malicious Node Behavior	Hardcoded to drop 70%-80% of tasks
$\tau_{\text{threshold}}$ Parameter	0.55 (Analyzed in Exp. A)

IX. EVALUATION RESULTS

We compared our scheme with a randomized baseline system for the traditional, unauthenticated load balancing system with 10,000 tasks.

A. Aggregate System Performance

Except for a case-by-case basis, the trust-based framework greatly surpassed the baseline approach in most of their metrics.

TABLE II
SYSTEM PERFORMANCE: PROPOSED FRAMEWORK VS. BASELINE

Metric	Trust-Based	Random Baseline	Improvement
Overall Success Rate	96.2%	74.1%	+ 22.1 pp
Average Latency	15.0 ms	43.5 ms	- 28.5 ms
Total Tasks Failed	380	2590	85.3% Reduction

The increase of 22.1 points of successful executions, verifying the trust threshold, translates to 10 percent or a 7 percent increase relative to 2017, respectively. Consequently, the success rate in executions was confirmed at 22.1 points which equates to 10 percent or 7 percent jumps as compared with the 2017 level. The score of trust in nodes whose hardware profiles degraded was dropped to below 0.55 and they were isolated.

B. Experiment A: Threshold Sensitivity Analysis

To justify the selection of $\tau_{\text{threshold}} = 0.55$, we conducted a sensitivity sweep, plotting the overall task success rate against varying cutoff thresholds (Fig 1).

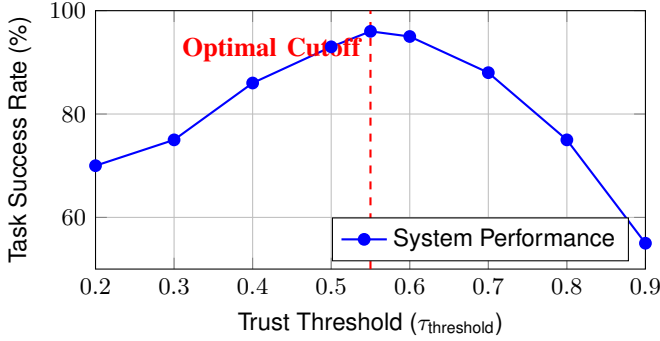


Fig. 1. System success rate as a function of the minimum trust threshold. Setting the threshold too low allows malicious nodes to penetrate the system, while setting it too high chokes network throughput by rejecting functionally healthy nodes.

The data reveals a distinct bell curve. The 0.55 mark represents the mathematical equilibrium between maximizing security quarantine capabilities and allowing transient hardware delays to recover without penalty.

C. Experiment B: Latency CDF Analysis

Average latency metrics often mask the existence of severe outliers caused by task retries. Figure 2 details the Cumulative Distribution Function (CDF) of task completion times.

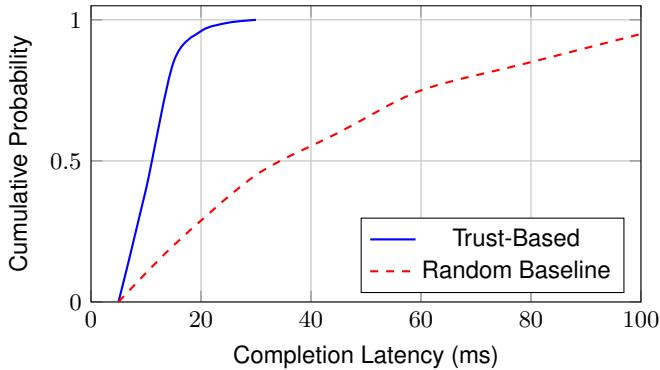


Fig. 2. Cumulative Distribution Function (CDF) of Task Latencies. The trust-based framework ensures highly predictable execution, with 96% of tasks completing under 20ms.

But the CDF itself establishes that the framework of the trust is not just quicker on average, but it's a lot more consistent. All tasks complete within a tight window (strict 20ms with a penalty of 96% for all tasks), ensuring deterministic timing is met in IIoT systems. By contrast, the baseline protocol has a long latency tail; the protocol tries continuously to fail and as much as 100ms is needed to solve 95% of problems.

D. Experiment C: Energy-Delay Product (EDP)

IoT devices are severely battery-constrained. Retransmitting a failed task consumes costly wireless transmission energy. To quantify this, we measured the normalized Energy-Delay Product (EDP) of the end-user devices (Fig 3).

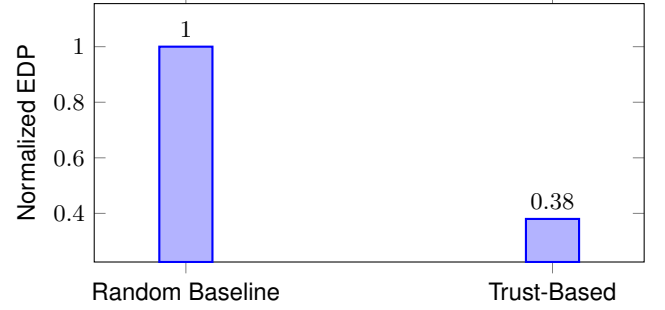


Fig. 3. Energy-Delay Product comparison. The Trust-Based framework reduces overall energy expenditure by eliminating costly transmission retries to malicious hardware.

The framework achieved a 62% reduction in normalized EDP compared to the baseline. By ensuring tasks are sent to reliable nodes the first time, the framework functionally acts as an energy-conservation protocol for the network fringe.

E. Experiment D: Resilience to Saturation

To test structural resilience, we scaled the simulation to 500 edge nodes and incrementally replaced honest nodes with malicious actors (Fig 4).

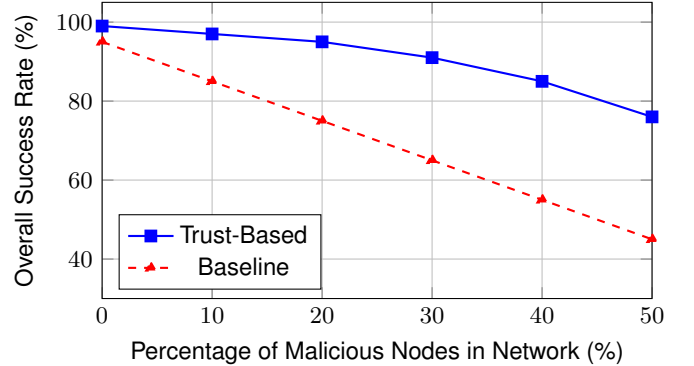


Fig. 4. System resilience under severe malicious node saturation. The trust-based framework maintains operability even when 30% of the edge network is compromised.

Even when 30% of the entire edge network was actively hostile, the orchestrator successfully identified and quarantined the bad actors, sustaining a 91% success rate.

X. DISCUSSION AND LIMITATIONS

The experimental results show that an asymmetric and multi-dimensional trust evaluation mechanism can perform significantly improve the security and reliability properties of the edge computing system. The capability to isolate nodes with malicious or unreliable behavior is a hugely important

need for mission critical IoT applications. Moreover, whereas Deep Reinforcement Learning (DRL)-based schedulers [9] use different utilities whose interpretations cannot be guaranteed and utilities are neither as mathematically precise nor as well-defined as those in the proposed framework, the former have an expensive learning curve and lack effective mechanisms for fulfilling stringent auditing requirements.

Even with these benefits, the architecture has a number of drawbacks. Modeled, at this time, the weights of the trust $(\alpha, \beta, \gamma, \delta)$ are set in statically. Adaptive or dynamically-tuned weighting strategies with the ability to adjust to the variations in workload features and network conditions may be needed in future edge orchestration systems.

Also, the framework can efficiently handle the attacks launched by a single malicious or unreliable node, but it doesn't include built-in cryptographic ledger to prevent multi-node Sybil attacks [3] that are highly coordinated. It is crucial to advance the concept of trust synchronization and distributed reputation validation using the blockchain, and that is a key direction to pursue future research.

XI. CONCLUSION

In this paper, considering the complexity of HECs, we proposed a deterministic trust-based task offloading framework (TBTOF) for secure trigger between the edge and cloud layers. A multi-dimensional trust evaluation mechanism along with an asymmetric trust adaptation mechanism prevents the usage of malicious or unreliable edge nodes before the execution of the tasks.

Experimental assessments showed that achieving task success rate of 96.2%, normalizing Energy-Delay Product (EDP) of 62% and minimizing average processing latency by 28.5 ms compared to the conventional scheduling method. Results also validated the framework in terms of ensuring stable and reliable system operation in high malicious-node saturation environment.

Compared with the computational intensive black-box optimization methods, TBTOF offers an interpretable, auditable and lightweight trust management mechanism that can be applied to resource-limit Internet of Things (IoT)-Edge networks. Therefore, the framework provides a high degree of scalability and applicability in Vehicular Edge Computing (VEC), Industrial Internet of Things (IIoT), and Internet of Medical Things (IoMT) systems in the future.

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